MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPES 5BP1A AND 5HP1A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Electrostatic deflection and focus

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	ed	Eb1	Eb2	Light output	Rg	Zd	Alt
Unit:	V	Vdc	v	Vdc	Vdc	ftL	Meg	Meg	ft
Maximum:	6.9	0	500	1,100	2,200	---	1.5	1.0	10,000
Minimum:	5.7	-125	---	---	1,500	2.0	---	---	---
TEST CONDITIONS:	6.3	Adj	---	Focus	1,500	---	---	---	---

GENERAL:

Qualification - Required

(A) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Qualification inspection						
5101	Neck and bulb alinement (electrostatic types)	-		Diam	---	1.87	Inches
Ⓐ 5216	Cathode illumination	-		---	---	---	---
Ⓐ 1216	Base material insulating quality	-	Zone 5 (minimum)	---	---	---	---
5248	Deflection factor uniformity	-		---	---	---	---
1331	Direct-interelectrode capacitance	-	Control grid to all	Cg1	---	12	pF
			D1 to D2	C1D2	---	3	pF
			D3 to D4	C3D4	---	3	pF
			D1 to all	CD1	---	15	pF
			D3 to all	CD3	---	15	pF
			D1 to all except D2	CD1	---	13	pF
			D2 to all except D1	CD2	---	13	pF
			D3 to all except D4	CD3	---	13	pF
			D4 to all except D3	CD4	---	12	pF
1141	Pressure (implosion)	-		---	---	---	---
5111	Vibration	-		Width	---	1.0	mm
	Quality conformance inspection, part 1						
5201	Voltage breakdown	-		---	---	---	---
5201	Voltage breakdown (electrostatic types)	-		---	---	---	---
5206	Gas "cross"	1	Light = 2 ftL	---	---	---	---
5106	Bulb, screen, and faceplate quality	-		---	---	---	---
5221	Light output	1		Light	2.0	---	ftL
5223	Modulation	-	Light = 2 ftL	ΔEc	---	35	Vdc
5231	Spot position (electrostatic deflection)	-		---	---	15	mm
5231	Spot displacement (leakage)	-		Displ	---	10	mm
5241	Grid cutoff voltage	-		Ec1	-15	-45	Vdc
5251	Grid No. 1 leakage current	-		---	---	---	---
5251	Anode No. 2 leakage current	-		---	---	---	---
	Quality conformance inspection, part 2						
1301	Heater current	-		If	540	660	mA
5201	Electrode current (anode No. 1)	-	Ec1 = 0	Ib1	-50	10	μAdc
Ⓐ 5201	Electrode current (cathode)	-	Light = 2 ftL	Ik	---	1,000	μAdc
5101	Base alinement (electrostatic types)	-	+3D4, pin No. 1	---	---	---	---
5101	Angle between traces	-		---	---	---	---
5101	Neck and base alinement (electrostatic types)	-		---	---	---	---
5216	Stray light emission (conventional types)	-	Eb2 = 2,200 Vdc	---	---	---	---

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shall be as specified in the contract or order

METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2 -Continued						
5226	Line width "A"	2	Light = 2 ftL	Width	---	0.65	mm
5226	Line width "B"	3	Light = 2 ftL	Width	---	0.65	mm
5246	Focusing voltage at cutoff	-		Eb1	265	400	Vdc
(A) 5246	Focusing voltage (zero bias)	-		Eb1	255	420	Vdc
5248	Deflection factor	-	1D2	DF	53	72	Vdc/in.
5248	Deflection factor	-	3D4	DF	48	66	Vdc/in.
(A) 1101	Secureness of base, cap, or insert	-		---	---	---	---
(A) 1111	Base pin solder depth	-		---	---	---	---
(A) 1105	Permanence of marking	-		---	---	---	---
	Quality conformance inspection, part 3						
---	Life test	-	Group C; Light = 2 ftL; Eb2 = 2,200 Vdc; t = 500 hours (minimum)	---	---	---	---
---	Life-test end points:						
5226	Line width "A"	-		Width	---	0.70	mm
5226	Line width "B"	-		Width	---	1.0	mm
5223	Modulation			ΔEc	---	35	Vdc

NOTES:

1. This test to be performed at the conclusion of the holding period.
2. The deflecting plates shall be returned to anode No. 2 through a minimum of 2.5 megohm resistors. The light output shall be set at 2 ftL. The high frequency scanning shall be applied to the deflecting plates nearest the screen and the amplitude shall be adjusted to give a line length of approximately 90 percent of the maximum tube diameter. The low frequency scanning amplitude shall be expanded to approximately 90 percent of the maximum tube diameter in the direction perpendicular to the direction of high frequency scanning. Readjustment may be made for best overall focus.

   (a) The tubes shall be observed for deflection defocusing, astigmatism, or spot ellipticity observable to the eye as evidenced by fuzziness due to lack of sharpness of trace (usually around edges) bow-tieing (irregular widths of any single line when observed at different points) bowing of trace other than that normally caused by curvature of bulb.

   (b) This test for focus is to be made in addition to the line width measurements.

3. The same conditions shall be set up as described in note 2 except that the connection of deflecting elements to the low and high frequency scanning supplies shall be interchanged and the amplitudes adjusted to 90 percent of the maximum tube diameter in both directions without any adjustment of focus from note 2 conditions. An examination for defocusing, astigmatism, or spot ellipticity shall be made as in note 2.

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Air Force - 11, 17, 80
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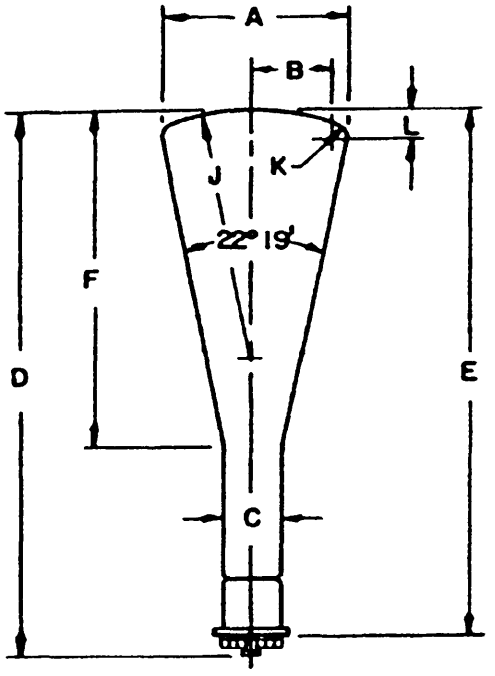
User activities:
Army - WC
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Preparing activity:
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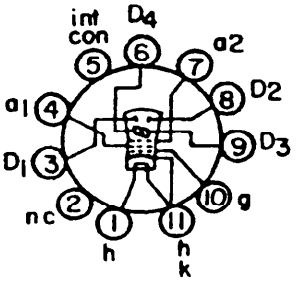
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	5.188 (131.78)	5.312 (134.92)
B	See note a	
C	1.562 (39.67)	1.688 (42.88)
D	16.375 (415.93)	17.125 (434.98)
Reference dimensions (see note d)		
E	16.188 (411.18)	
F	10.250 (260.35)	
J	8.000 (203.20) R	
K	.500 (12.70) R	
L	.808 (20.52)	



NOTES:
a. The minimum useful screen radius shall not be less than 2.25 inches (57.15 mm).
b. The base for 5BP1A shall be a large wafer magnal 11-pin with medium straight sleeve (B11-35) type, or medium magnal 11-pin shell type (B11-66). The base for the 5HP1A shall be a large micanol-wafer magnal 11-pin with medium straight sleeve (B11-35) type, or medium magnal 11-pin low-loss shell type (B11-66).
c. The bulb shall be a J42C type.
d. Reference dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing for electron tube types 5BP1A and 5BHP1A.

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